

Aidan Shilling

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EDUCATION

University of Edinburgh

Master of Science, Computer Science

- Dissertation: Integrating Qurts Sub-structural Automatic Uncomputation with MLIR

Sep 2025 — Aug 2026

Edinburgh, UK

Drexel University

Bachelor's of Science, Computer Science

Sep 2019 — Aug 2023

Philadelphia, PA, USA

PROFESSIONAL EXPERIENCE

AMD

Product Engineer 2, ATE Infrastructure

- **Spotlight Award:** Led development of test infrastructure for server silicon; recognized for a database and query fix that prevented an estimated \$41 million in quarterly test time losses
- Maintained and extended a Ruby-based transpiler framework that compiled a unified test program representation to multiple ATE platform targets, producing hundreds of releases across 20+ silicon modules
- Owned CI/CD pipelines serving 40 engineers across 20+ per-module sub-pipelines, supporting 50-100+ daily test runs for automated validation of server silicon test software
- Designed and built a custom VCS in Rust for internal characterization database files, using an immutable append-only model for safe concurrent access across automated tooling, reducing workflow overhead from days to minutes

Sep 2023 — Sep 2025

Austin, TX, USA

AMD

Product Engineering Architect Co-op

- Developed and deployed a compiler in Rust, implementing a PEG parser & AST lowering for external IP test pattern code into AMD's internal language, reducing generation time by 80%
- Invited to give a live talk at an internal Tech Showcase on applying compiler design to solve engineering problems

Sep 2021 — Aug 2022

Austin, TX, USA

Imalogix

Software Engineer Co-op

- Owned end-to-end development of 10-15 report types for a healthcare SaaS platform used by hospital radiation technicians, building REST APIs, SQL queries, and frontend features across .NET MVC, jQuery, and Vue
- Built backend data aggregation pipelines in SQL and C# combining outputs from multiple sources including in-house computer vision models, delivering unified report data to frontend consumers

Sep 2020 — Mar 2021

Bryn Mawr, PA, USA

PROJECTS

Qurts-core Compiler Frontend, MSc Dissertation

- Designing a custom MLIR dialect in Rust preserving lifetime-parameterised substructural types as first-class IR constructs for a quantum programming language, with a PEG parser (PEST) and lowering pass from the Qurts-core AST to LLVM IR

Sep 2025 — Present

LLVM AMDGPU Backend, Open Source Contributor (github.com/llvm/llvm-project)

- Contributed `GlobalISel` `RegBankLegalize` rules for `amdgcns_alloc_vgpr`, `amdgcns_sat_pk4_i4_i8`, and `amdgcns_sat_pk4_u4_u8` intrinsics to the AMDGPU backend in LLVM (192937, 193096)

2026

Distributed Primality Checker, UoE Coursework

- Built a distributed system in Go with a coordinator, workers, and custom filesystem communicating via gRPC and Protocol Buffers, with logical timestamp tracking and result caching

2026

Pingulon.net, Personal Project (pingulon.net)

- Built and self-hosted a personal short story publishing platform using SvelteKit and a Go REST API backed by MongoDB, containerized with Docker

2026

SKILLS

- **Programming:** Rust, Go, Python, JavaScript, C/C++, Ruby, SQL, Functional Programming
- **Technologies:** MLIR, LLVM, gRPC, Protocol Buffers, Ruby on Rails, React.js, AWS, Git, Linux/Unix